

# HRI Survey

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## Start of Block: Consent Form

Human Perception of Robot Behaviour IRB ref no. IRB-2025-765. You are invited to participate in a research study on human perception of a robot's mental state based on its expressive movements. This study is conducted by [REDACTED]. This study will take approximately 10-12 minutes of your time. In this online survey, you will view a series of videos where a single arm robot will perform different expressive movements associated with different mental states. The questions will pertain to your recognition of the type and degree of the robot's mental state. Your participation in this study is completely voluntary, and you have the right to withdraw from the study at any time without any penalty. If you do not wish to complete this survey, just close your browser. Your participation in this research will be kept confidential, and data will be averaged and reported in aggregate. Your responses will be anonymous and IP addresses will not be collected to guarantee complete anonymity. Possible outlet of dissemination may be research articles. Although your participation in this research may not benefit you personally, it will help us to incorporate seamless nonverbal communication between humans and robots, thereby improving human-robot interaction in healthcare, education, manufacturing, entertainment and other such industries. There will be no compensation provided. There are no risks to individuals participating in this survey beyond those that exist in daily life. If you have questions about this project, you may contact [REDACTED]. This project has been reviewed and approved by Institutional Review Board. Questions concerning your rights as a participant in this research may be directed to the IRB at IRB@[REDACTED] or call 6592 2495. Please print a copy of this consent form for your records, if you so desire. I have read and understood the above consent form.

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## Consent Form

☐ I have read and understood the above consent form. I certify that I am 18 years old or older and, by clicking the next button to enter the survey, I indicate my willingness to voluntarily take part in the study.

☐ I do not wish to participate in this study and end the survey by clicking on the next button.

End of Block: Consent Form

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Start of Block: End of Survey if consent rejected

Thank you for your time. You have chosen not to participate in this study. Your response has not been recorded, and you may now close this window.

End of Block: End of Survey if consent rejected

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Start of Block: Background Information

## Section 1: Demographic Information



What is your gender?

- ☐ Male
- ☐ Female
- ☐ Other
- ☐ Prefer not to say



What is your age?

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What ethnicity would you describe yourself as?

- ☐ Chinese
- ☐ Malay
- ☐ Indian
- ☐ Eurasian
- ☐ Other \_\_\_\_\_
- ☐ Prefer not to say

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What is the highest degree or level of school you have completed?

- ☐ Less than a high school diploma
- ☐ High school degree or equivalent
- ☐ Bachelor's degree or equivalent
- ☐ Master's degree or equivalent
- ☐ Doctorate or professional degree

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How familiar are you with robots?

|             | Not Familiar          | Slightly Familiar     | Moderately Familiar   | Very Familiar         | Extremely Familiar    |
|-------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Familiarity | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

How comfortable are you in working with robots?

|         | Not<br>Comfortable    | Slightly<br>Comfortable | Moderately<br>Comfortable | Very<br>Comfortable   | Extremely<br>Comfortable |
|---------|-----------------------|-------------------------|---------------------------|-----------------------|--------------------------|
| Comfort | <input type="radio"/> | <input type="radio"/>   | <input type="radio"/>     | <input type="radio"/> | <input type="radio"/>    |

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Page Break

## Section 2: Analyzing Robot Expressive Movements and Mental State

In this section, you will see a set of videos showing different expressive movements of robots.

Observe each expression closely

<Video: 4States\_Comparison\_Unified>

For each expression, what words would you use to describe the mental state of the robot?

- ☐ Expression A \_\_\_\_\_
- ☐ Expression B \_\_\_\_\_
- ☐ Expression C \_\_\_\_\_
- ☐ Expression D \_\_\_\_\_

JS

Select the mental state of the robot that best matches **Expression A**

▼ Confidence ... None

Display this question:

If Select the mental state of the robot that best matches Expression A = Other

If other, then please enter the mental state that best matches **Expression A**

\_\_\_\_\_

Display this question:

*If Select the mental state of the robot that best matches Expression A = Confidence*

*Or Select the mental state of the robot that best matches Expression A = Curiosity*

*Or Select the mental state of the robot that best matches Expression A = Fear*

*Or Select the mental state of the robot that best matches Expression A = Hesitation*

*Or Or If other, then please enter the mental state that best matches Expression A Text Response Is Not Empty*

JS

Rate the degree of **(no selection yet)** in Expression A, with 1 being slightly (no selection yet) and 5 being extremely (no selection yet).

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

JS

Select the mental state of the robot that best matches **Expression B**

▼ Confidence ... None

Display this question:

*If Select the mental state of the robot that best matches Expression B = Other*

If other, then please enter the mental state that best matches **Expression B**

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Display this question:

*If Select the mental state of the robot that best matches Expression B = Confidence*

*Or Select the mental state of the robot that best matches Expression B = Curiosity*

*Or Select the mental state of the robot that best matches Expression B = Fear*

*Or Select the mental state of the robot that best matches Expression B = Hesitation*

*Or Or If other, then please enter the mental state that best matches Expression B Text Response Is Not Empty*

JS

Rate the degree of **(no selection yet)** in Expression B, with 1 being slightly (no selection yet) and 5 being extremely (no selection yet).

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

JS

Select the mental state of the robot that best matches **Expression C**

▼ Confidence ... None

Display this question:

*If Select the mental state of the robot that best matches Expression C = Other*

If other, then please enter the mental state that best matches **Expression C**

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Display this question:

*If Select the mental state of the robot that best matches Expression C = Confidence*

*Or Select the mental state of the robot that best matches Expression C = Curiosity*

*Or Select the mental state of the robot that best matches Expression C = Fear*

*Or Select the mental state of the robot that best matches Expression C = Hesitation*

*Or Or If other, then please enter the mental state that best matches Expression C Text Response Is Not Empty*

JS

Rate the degree of **(no selection yet)** in Expression C, with 1 being slightly (no selection yet) and 5 being extremely (no selection yet).

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

JS

Select the mental state of the robot that best matches **Expression D**

▼ Confidence ... None

Display this question:

*If Select the mental state of the robot that best matches Expression D = Other*

If other, then please enter the mental state that best matches **Expression D**

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Display this question:

*If Select the mental state of the robot that best matches Expression D = Confidence*

*Or Select the mental state of the robot that best matches Expression D = Curiosity*

*Or Select the mental state of the robot that best matches Expression D = Fear*

*Or Select the mental state of the robot that best matches Expression D = Hesitation*

*Or Or If other, then please enter the mental state that best matches Expression D Text Response Is Not Empty*

JS

Rate the degree of **(no selection yet)** in Expression D, with 1 being slightly (no selection yet) and 5 being extremely (no selection yet).

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

End of Block: Background Information

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Start of Block: Block 5

Now that you've matched a mental state to each robot expression, let's compare slight variations of those expressions. For each question, rate how strongly the robot shows that mental state on a scale from 0 to 4, where 0 means no sign of the mental state and 4 means it's displayed very strongly.

End of Block: Block 5

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Start of Block: Compare: Confidence

<Video: Comp\_CON\_DOP\_12\_Unified>

Rate the degree of **Confidence** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

<Video: Comp\_CON\_SOA\_21\_Unified>

Rate the degree of **Confidence** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

End of Block: Compare: Confidence

Start of Block: Compare: Fear

<Video: Comp\_FEAR\_DOP\_21\_Unified>

Rate the degree of **Fear** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

<Video: COMP\_FEAR\_DOS\_21\_Unified>

Rate the degree of **Fear** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

< Video: Comp\_FEAR\_NOP\_12\_Unified>

Rate the degree of **Fear** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

<Video: Comp\_FEAR\_NOR\_12\_Unified>

Rate the degree of **Fear** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

< Video: COMP\_FEAR\_SOA\_21\_Unified>

Rate the degree of **Fear** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

<Video: COMP\_FEAR\_SOR\_12\_Unified\_V2>

Rate the degree of **Fear** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

End of Block: Compare: Fear

Start of Block: check



Please enter your birth year

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End of Block: check

Start of Block: Compare: Curious

<Video: Comp\_CUR\_DOP\_12\_Unified>

Rate the degree of **Curiosity** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

<Video: COMP\_CUR\_DOT\_12\_Unified>

Rate the degree of **Curiosity** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

<Video: COMP\_CUR\_LAH\_21\_Unified>

Rate the degree of **Curiosity** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

<Video: COMP\_CUR\_LAV\_21\_Unified>

Rate the degree of **Curiosity** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

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<Video: Comp\_CUR\_SOA\_21\_Unified>

Rate the degree of **Curiosity** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

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<Video: Comp\_CUR\_SOT\_21\_Unified\_V2>

Rate the degree of **Curiosity** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

End of Block: Compare: Curious

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Start of Block: Compare: Hesitant

<Video: Comp\_HES\_DOP\_21\_Unified>

Rate the degree of **Hesitation** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

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<Video: Comp\_HES\_DOR\_21\_Unified>

Rate the degree of **Hesitation** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

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<Video: Comp\_HES\_DOT\_12\_Unified\_V2>

Rate the degree of **Hesitation** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

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<Video: Comp\_HES\_NOR\_12\_Unified>

Rate the degree of **Hesitation** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

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<Video: Comp\_HES\_SOA\_12\_Unified>

Rate the degree of **Hesitation** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

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<Video: Comp\_HES\_SOR\_12\_Unified\_V2>

Rate the degree of **Hesitation** displayed by Robot 1 and Robot 2

|         | 0                     | 1                     | 2                     | 3                     | 4                     |
|---------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Robot 1 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Robot 2 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

End of Block: Compare: Hesitant

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